

Analysis First-Year Exam, August 2016, Part 1

Answer exactly *FOUR* questions. Write solutions in a neat and logical fashion, giving complete reasons for all steps.

1. Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be bounded real sequences and let $(c_n)_{n=1}^{\infty}$ be the sequence

$$a_1, b_1, a_2, b_2, \dots$$

Prove that

$$\limsup_{n \rightarrow \infty} c_n \leq \max \left(\limsup_{n \rightarrow \infty} a_n, \limsup_{n \rightarrow \infty} b_n \right)$$

and that if $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ converge to γ , then so does $(c_n)_{n=1}^{\infty}$.

2. State the definition of a perfect set and prove that the standard Cantor set in $[0, 1]$ is a perfect set.
3. Let $f : X \rightarrow Y$ be a map between metric spaces and let

$$G_f = \{(x, f(x)) : x \in X\} \subseteq X \times Y$$

be its graph. Prove that if G_f is compact then f is continuous.

4. Let f be a uniformly continuous real-valued function on the bounded open interval (a, b) . Prove that the real limit $\lim_{t \rightarrow b^-} f(t)$ exists. Show by example that we may not replace uniformly by bounded without jeopardizing this conclusion.
5. Let the real-valued function f be continuous on $[0, 2]$ and twice-differentiable on $(0, 2)$. Assume that $f(0) = 0$, that $f(1) = 1$, and that $f(2) = 2$. Prove that there exists $u \in (0, 2)$ such that $f''(u) = 0$.